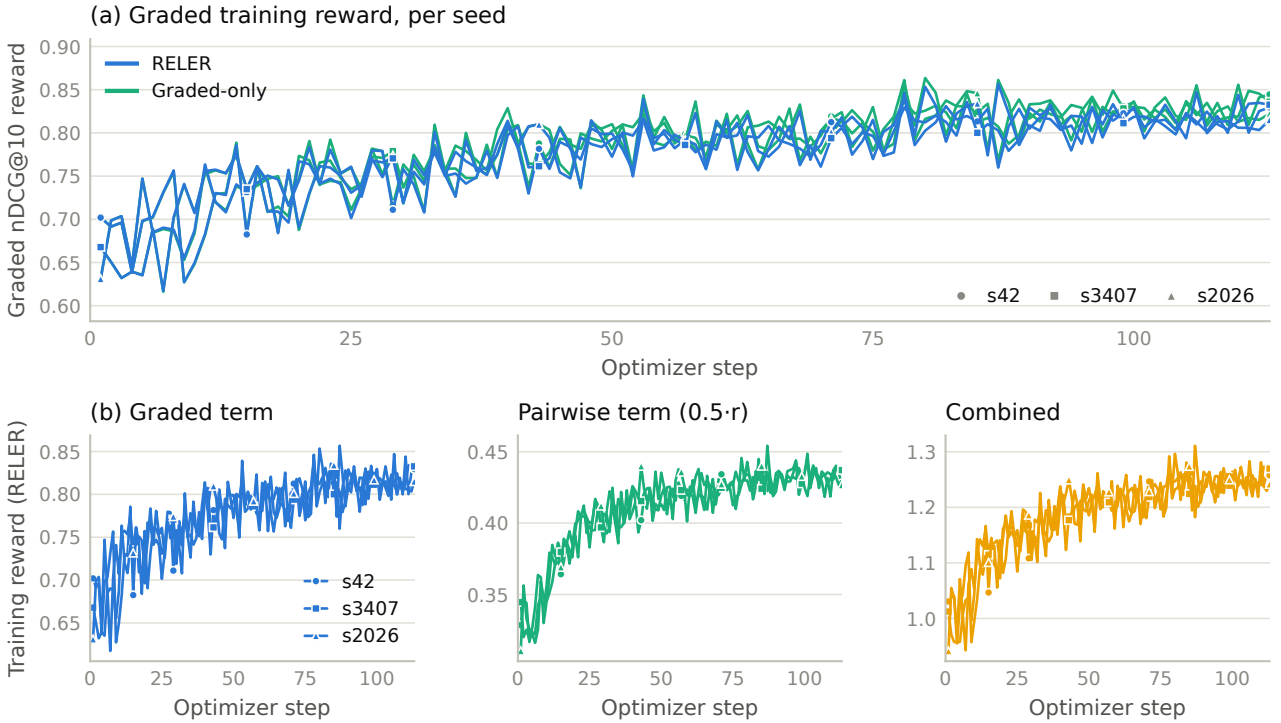
(a) Graded training reward, per seed
Graded nDCG@10 reward
RELER
Graded-only
s42
s3407
s2026
Optimizer step
(b) Graded term
Training reward (RELER)
Pairwise term (0.5·r)
Combined